

An In-Tool Recommender System for Computer Graphics Education with Real-Time Learner Profiling

Alberto Cannavò, *Member, IEEE*, Roberta Macaluso *Member, IEEE*,
Pier Luigi Nakai Ricchetti, Fabrizio Lamberti *Senior Member, IEEE*

Abstract—Supervising large cohorts during practical laboratory sessions is a major challenge in technical education, especially in Computer Graphics (CG) courses, where learners must master steep-learning-curve 3D suites such as Blender, Maya, etc. The procedural flexibility of these tools, allowing the same outcome through multiple workflows, makes it hard for instructors to trace learners’ difficulties to their root cause. Educational recommender systems (ERSs) can help by identifying struggling learners and tailoring resources to individual needs. **However, ERSs still face barriers such as difficulty adapting recommendations to learners’ evolving needs and limited usability due to interfaces detached from the learning environment.** Moreover, CG remains underexplored compared to other academic domains, and no validated system has been natively integrated into a professional 3D authoring tool for continuous tracking and timely support. To address this gap, this paper introduces an ERS integrated into a well-known 3D graphics tool that monitors learner interactions during laboratory activities related to a CG course, incrementally builds a profile from observed difficulties, provides immediate and continuous feedback on task execution, and suggests the most relevant tutorials from a library using a content-based filtering strategy. The proposed approach combines a TF-IDF-based tutorial representation, a revised Rocchio update mechanism for continuous profiling, and cosine similarity for ranking candidate tutorials. An experimental evaluation was carried out by involving 26 participants to assess effectiveness, usability, and perceived learning support in a representative educational setting. Both objective and subjective metrics confirmed the benefits brought by the use of the proposed ERS.

Index Terms—Educational recommender systems, computer graphics education, Blender, content-based filtering, continuous profiling, immediate feedback, tutorial recommendation.

I. INTRODUCTION

PRACTICAL laboratory sessions play an important role in technical education, engaging learners in hands-on tasks under expert guidance [1]–[3]. Nevertheless, supervising large cohorts simultaneously remains a significant challenge [4], [5]. This challenge is particularly critical in Computer Graphics (CG) courses, where learners are requested to master 3D spatial reasoning and complex, steep-learning-curve graphics suites such as Blender, Maya, etc. [6]–[8]. The literature on CG education highlights how novice learners frequently struggle with the inherent procedural flexibility of these environments, since, in such suites, the same outcome can often

be achieved through multiple distinct workflows [9]. These alternative workflows hinder instructors from tracing learner difficulties back to their underlying conceptual or procedural origin and this challenge is further amplified when instructors support multiple learners progressing at different paces. In this respect, educational recommender systems (ERSs) [10] can offer practical solutions to directly address these challenges by enabling analytics-driven identification of struggling learners, delivering immediate and continuous targeted assistance during laboratory sessions [11], [12], and tailoring learning resources to individual prior knowledge, skills, and goals [13], [14]. Empirical studies confirm that ERSs can improve academic performance, engagement, and motivation [12], [14], [15], and enhance learners’ autonomy and critical thinking [16]–[18].

Despite these benefits, ERSs still face technical and adoption barriers. For example, matching recommendations to diverse learner needs is challenging, as mismatches between user expectations and recommendations are common [19]; addressing this issue requires dynamic modeling strategies that capture learners’ progress over time, enabling recommendations to be adjusted continuously rather than computed once and left static.

A further obstacle is usability and trust: learners and instructors often do not understand why recommendations change or how to act on them, since explanations and interfaces are often inadequate, thus affecting acceptance and use of these solutions [20], [21]. Addressing this limitation requires transparent, user-centered interfaces embedded directly into the environments where learning naturally occurs, rather than delivered through separate dashboards and tools.

In addition, although in the past decade the use of ERSs has seen a significant increase in traditional academic domains, such as mathematics, physics, and computer science, a notable research gap still remains in specialized creative and technical fields, particularly CG, as pointed out by recent work such as [22]–[24]. Moreover, to the best of the authors’ knowledge, no validated system capable of tracking learners’ actions and offering timely support has been natively integrated into a professional 3D authoring tool for educational purposes. This gap suggests the need for novel approaches that embed continuous monitoring and immediate feedback, and adaptive recommendation capabilities directly within the software environments used in CG courses.

Considering the above aspects, this paper presents a novel ERS for the CG domain integrated within a graphics suite—namely Blender—that can be leveraged to support learners during practical laboratory sessions. Blender was chosen since

Alberto Cannavò, Roberta Macaluso and Fabrizio Lamberti are with the GRAINS – GRaphics And INtelligent Systems group at the Dipartimento di Automatica e Informatica di Politecnico di Torino, 10129 Torino, Italy. E-mail: {alberto.cannavo, roberta.macaluso, fabrizio.lamberti}@polito.it.

Pier Luigi Nakai Ricchetti is with Navigare, São Paulo, Brazil. Email: ricchetti@nvgr.io.

Manuscript received Xxxxx XX, XXXX; revised Xxxxx XX, XXXX.

it is a widely adopted, professional-grade suite increasingly employed for educational purposes (as also free and open source) [9], [25]–[27]. Operating directly within Blender, the system compares each operation performed by the learner against the expected correct sequence for the exercise at hand. **Whenever a variation is detected, it updates the learner profile and immediately reflects this in the interface, providing continuous feedback about whether the current step was performed correctly or contains errors to fix, while the exercise is still in progress.**

Building on this monitoring and profiling capability, the system is then capable of recommending targeted tutorials via content-based filtering, thus matching tutorial content with the learner’s accumulated difficulties. More specifically, the system adopts the Term Frequency – Inverse Document Frequency (TF-IDF) technique for the vectorization of tutorial content, a revised version of the Rocchio algorithm to update continuously the learner profile and describe it as a vector in the same space as the tutorial, and cosine similarity for matching the two closest vectors [28], [29]. **The code of the ERS is available for download at <https://tinyurl.com/2zzzf2ns>.** To design and validate the devised approach, an actual introductory CG course taught at the authors’ university¹ was considered as a case study.

The remainder of this paper is organized as follows. Section II reviews the related work. Section III introduces the educational context and motivates the proposed approach. Section IV presents the overall system architecture, while Section V details the recommender algorithm. Section VI describes the protocol adopted for the experimental evaluation. Section VII presents the results, which are then discussed in Section VIII. Finally, Section IX concludes the paper.

II. RELATED WORK

Recent ERS research has increasingly explored deep learning (DL) and reinforcement learning (RL) approaches [30]. For instance, the authors of [31] surveyed DL-based recommender systems for e-learning, highlighting their ability to learn complex user and item representations, as well as temporal dependencies, directly from data. However, the survey also pointed to the challenges of training such models when interaction data are sparse and computational resources are limited. Similar challenges have also emerged when considering RL-based solutions, as they remain data-hungry, computationally expensive, and often difficult to interpret, as reported in [32]. Moreover, most RL approaches rely on a closed interaction loop, where weights are optimized through numerical rewards assigned after each action. This paradigm introduces additional challenges, including the difficulty of designing an appropriate reward function and sample inefficiency, as these methods typically require extensive interaction data to learn effective representations [32]. These issues are particularly critical in educational domains characterized by complex, multidimensional learning objectives that are not easily reducible to simple quantitative feedback [32].

Building on these considerations, this work deliberately avoids such computationally demanding black-box models and proposes an ERS able to operate with limited data and under the resource constraints of laboratory settings. To this aim, the proposed system leverages the advantages offered by the content-based filtering (CBF) approach as recommendations can be generated from item content without requiring a large collection of historical user-item interactions [33], [34]. This property is essential in educational settings where interaction logs are typically sparse [35]. Other recommendation paradigms impose different requirements that may be less suitable for this setting. Collaborative filtering (CF) methods, for instance, generally perform well only in the presence of dense interaction data and suffer from domain shift [36]. Knowledge-based (KB) approaches, although powerful in modeling long-term or infrequently expressed preferences, often require explicit user input and careful modeling of relationships among users, items, and learning goals [37]. CBF therefore provides a lightweight alternative that can exploit the content of the available learning resources while progressively adapting recommendations as information about the learner is collected.

While the previous discussion motivates the choice of a lightweight and interpretable CBF approach, a second and orthogonal issue concerns the application domain. As anticipated in Section I, the broader education literature reports that domain-specific ERSs for areas such as CG are still underexplored compared with more traditional academic domains [38]. As a result of the limited research conducted in this area, the literature lacks ERSs directly integrated into professional CG authoring tools, despite the potential benefits of such integration in terms of reduced workflow friction, richer interaction data, and improved perceived usefulness and trust [39].

The few notable examples in the CG domain mainly emphasize tools and pedagogy rather than proposing personalized, in-tool recommendation systems. For instance, the work in [7] categorizes CG teaching tools and potential challenges (e.g., scaffolding practical skills, authentic tool use). However, this study does not account for ERSs integrated directly within the workflow of tools used to produce graphics content. The authors of [6] promote Blender as an accessible and versatile tool for simulation and 3D printing in early STEM education, without considering the possibility of supporting adaptive guidance or user profiling within this software. Similarly, the work in [40] discusses experimental methods aimed at enhancing the creative potential of higher education students using modern design software (including Blender), again without incorporating an ERS. Taken together, these studies demonstrate the relevance of professional CG software in education, but leave open the question of how learner behavior can be continuously analyzed and used to provide personalized support directly within the authoring environment.

This work specifically addresses the above research gaps by proposing an ERS integrated into Blender to support hands-on 3D modeling activities of an actual introductory CG course, combining continuous learner profiling, immediate feedback, and content-based tutorial recommendation. The Blender add-

¹Computer Graphics course description: <https://tinyurl.com/4ct9hewy>

on proposed in this work follows the same integration principles suggested in [39], but embeds the recommendation system directly within a professional 3D tool rather than within a generic Learning Management System (i.e., Moodle). In summary, to the best of the authors' knowledge, the proposed system is the first domain-specific ERS for CG integrated within a well-known 3D graphics suite.

III. CONTEXT AND MOTIVATIONS

As anticipated, an actual CG course served as the educational context that informed the design of the proposed approach. In the considered course, learners perform hands-on 3D modeling exercises using Blender, consult learning materials when difficulties arise, and then apply the acquired knowledge to new exercises requiring similar operations. Ideally, an instructor would attempt to address learner difficulties through one-on-one support. However, in large classes, it is often impractical for a single instructor (or even a small team of collaborators) to provide timely, individualized feedback to every learner simultaneously. Consequently, the common practice is for the instructor to move between laboratory workstations, identifying learners who appear stuck, offering guidance to resolve specific issues, and recommending tutorials to address any remaining knowledge gaps.

Although instructors may be able to easily identify the problem, they could struggle to recommend the most relevant resources for the particular learner because they typically observe only part of the modeling process rather than the complete sequence of actions leading to a difficulty. This limited insight into the learner's actions is particularly problematic with modeling tools, since the same result (or mistake) can be obtained in many different ways, making the final output alone potentially misleading. As a result, learners may need to independently search the available learning materials to diagnose the problem and identify the knowledge or skills required to overcome it. This approach is not always effective, particularly when learners lack sufficient insight into the specific sequence of actions that led to the encountered difficulty. Moreover, the process of identifying the most appropriate tutorials can be time-consuming and cognitively demanding. *These limitations motivate the need for a system that monitors learner actions to provide two types of support: i) immediate and continuous feedback during the execution of the modeling exercise increases the learner's awareness of their current completion status and correctness; and ii) targeted learning recommendations upon completion to address difficulties automatically identified by the system.*

The course was also considered as the case study for validating the proposed approach. More specifically, it served as the source of the learning materials used during the experiment, established the participant selection criteria by requiring prior attendance of a similar Blender-based modeling course, informed the design of the experimental protocol and workflow phases, and provided the reference laboratory approach adopted as the baseline for comparison with learners supported by the proposed system.

IV. SYSTEM OVERVIEW

The proposed ERS, integrated within Blender, supports learners in automatically identifying operations that cause difficulties, providing immediate and continuous feedback that can enhance awareness of their weaknesses during the execution of an exercise or a tutorial. Once an exercise or tutorial has been completed, the ERS can then recommend personalized tutorials from an existing library to address the learner's specific difficulties. This library includes tutorials generated by instructors using the ERS.

To support the mentioned functionalities, the system was designed to allow users (when not specified, the term user refers to both learners and instructors) to operate in two modalities:

- *Log mode*: this modality is used by instructors to create tutorials by performing the target workflow in Blender. The system records the resulting sequence of structured operations performed by the instructor as a log file, which serves as a reference workflow for the exercise. Once created, the tutorial is added to the library and made available for recommendation to learners.
- *Tutorial mode*: this modality is used by learners to perform a tutorial from the library, previously generated in Log mode. The system loads the selected tutorial as the reference sequence, monitors the learner's actions, validates each operation against the expected workflow, *provides immediate and continuous feedback after each action performed by the learner, and recommends tutorials upon completion to address identified difficulties.*

The system is embedded into Blender as an add-on and the core components of its architecture (shown in Fig. 1) are described below.

A. Interaction Monitor

The Interaction Monitor component is designed to continuously monitor user input and events, such as key presses, mouse movements, or interactions with the GUI. The recognition of events is handled through the Blender APIs². Once events are detected, details of the operation performed as well as the context in which the operations took place are propagated to other components of the architecture in order to process and validate them according to the current modality (as shown in Fig. 1).

B. Operation Processor

This component translates the operation detected by the system into a structured representation that can be used by the other components of the architecture. Translation is required since the representation of the operations in Blender does not adhere to a specific standard format, thus resulting in each operation having different property names, values, organizations, and variable types. The proposed structured representation was designed to be flexible enough to accommodate different types of Blender's variables, data structures, and property

²Blender APIs <https://docs.blender.org/api/5.0/>

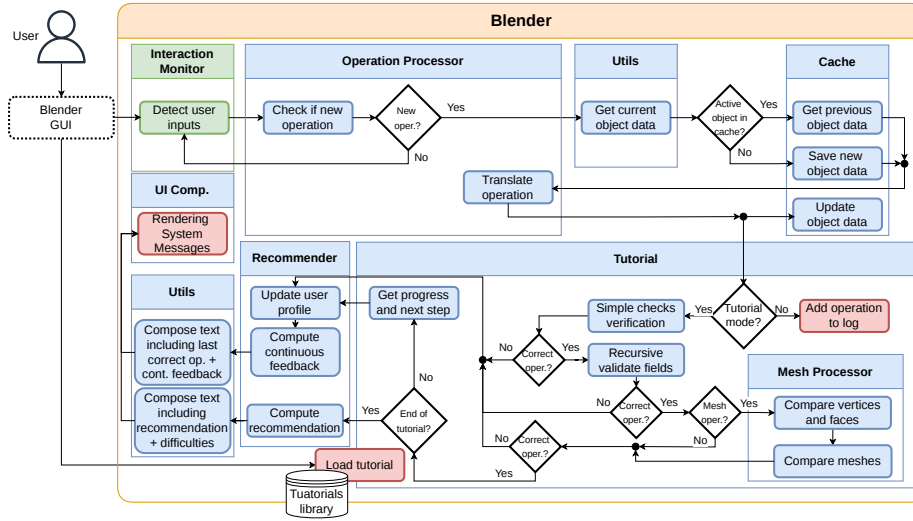


Fig. 1. Overall system architecture and interaction process in Blender. Green and red blocks denote the entry and exit points of execution loops, respectively.

descriptions while maintaining, at the same time, a fixed structure that could guide further manipulations of the other components. More specifically, this representation includes:

- *operation name*: obtained by querying Blender’s API;
- *operation properties*: containing all the properties of the operation, processed and organized in a dictionary where the keys are the property names that map to their respective values;
- *target object*: specifying the object on which the action is performed;
- *additional properties*: containing additional properties of the operation (organized in a dictionary) that are used to filter properties and validate tutorial steps, as detailed in the following sub-sections.

For instance, Fig. 2 provides an example of how the system translates an operation for modifying a specific parameter (i.e., setting the ambient light intensity to 0.5) into the structured ERS representation.

The translation rules were derived from empirical observations to first understand Blender’s underlying encoding structure for each operation, which was subsequently confirmed and expanded using the Blender API documentation.

The comprehensive processing envisioned for the translation ensures that the operation log is extremely detailed and highly informative for the following components of the system. As a matter of example, the component can manage users’ actions that are not internally recognized as operations by Blender, such as changing tabs, selecting a component in the UI, etc. Moreover, the component is capable of recognizing whether the detected action corresponds to a newly performed operation or merely a spurious event. This distinction is required since inputs such as clicks may not always indicate a completed operation (e.g., a left-click in an empty area).

After an operation is translated, it is also essential to capture and store the current state of the 3D objects in the scene. This need arises from the fact that the Recommender component must rely on an updated learner profile that is as accurate as

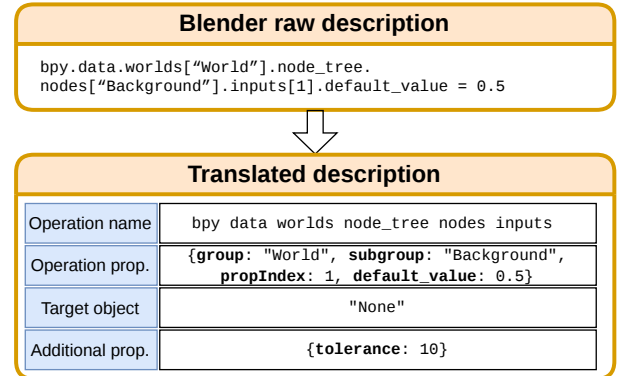


Fig. 2. Example of translating Blender’s internal representation of a specific operation into the structured ERS representation. The considered operation sets the ambient light intensity to 0.5, identified as property index 1 (*propIndex: 1*) of the background lighting module (the *Background* node) within the *World* environment settings. A *Target object* defined as “None” indicates that the operation directly acts on the same object without affecting others. The *tolerance* parameter listed among the *additional properties* specifies an acceptable percentage threshold used by the system when comparing the numerical value of the ambient light intensity to validate the operation.

possible in order to offer relevant recommendations. However, in the context of 3D modeling, there are often multiple ways to achieve a particular goal, and the system must be able to recognize and take these different approaches into account. Therefore, the Processor component interacts with the Utils and Cache components to retrieve and update data available for all the objects, such as vertices and faces with their respective 3D coordinates, transformations applied, operating mode used (whole object or mesh), etc.

In this way, it is possible to detect, e.g., whether the user has added, removed, or transformed a 3D object or any of its vertices, as well as to determine which elements are currently selected or which is the new name the user assigned to the object. All these detected modifications are included

in the additional properties field of the structured operation representation. This detailed information aids the system in accurately updating the learner profile and understanding the nuances of performed actions.

C. Tutorial

As shown in Fig. 1, the behavior of the Tutorial component depends on whether the system is running in Log or Tutorial mode.

When in Log mode, this component is responsible for creating and continuously updating a log file that records all operations performed by the instructor within the Blender GUI. This process continues until the instructor terminates the execution by interacting with the dedicated GUI element. Once the execution is terminated, the log file includes all the operations already processed by the Operation Processor component into the structured format described above. Using the same format internally adopted by the system to encode operations makes it possible to use the generated log file directly as a tutorial without any additional processing. The generated tutorial is included in the tutorial library and can be recommended by the system when operating in Tutorial mode.

Most of the functionalities of this component, though, are available when the system is in Tutorial mode. The first functionality is loading a tutorial file from the library. The tutorial name can be specified using the system GUI. Each line read from the file corresponds to an operation to be performed, i.e., a single step of the tutorial. Once a tutorial is loaded, the component validates the operations performed by the user against those that should have been performed according to the current step of the tutorial, as shown in the workflow depicted in Fig. 1. Since, as said, there may be various ways to achieve the same result, the validation process must be sophisticated enough to avoid detecting false errors or false successes, which could adversely affect the system's recommendations.

To reduce the system's overhead in the validation, the component first performs a number of simple checks, i.e., it evaluates conditions that can be unambiguously considered as wrong given certain conditions, such as verifying if the user is working in the correct mode (Object vs Edit Mode) and with the expected target object. Then, the component makes use of more sophisticated methods to validate the properties of the operation.

More specifically, for the validation of complex operations, the component relies on a recursive algorithm capable of accessing all properties of the performed operation, even when nested properties exist. It compares the current values of the operation's properties with those in the target one, i.e., those described in the tutorial file, allowing for an error tolerance (expressed as a percentage). This tolerance can be defined during the creation of the tutorial, and it has a default value set to 10%.

However, the expected result of a complex operation can sometimes be achieved by performing a given sub-operation a different number of times than specified in the tutorial, or by executing the sub-operations in a different order. In such cases, the validation method must be able to recognize the

user-performed operation as correct even if it differs from the one specified in the tutorial, as long as the final result is the same.

The approach adopted to deal with this need was to allow the system to compare not only the current tutorial step but also a series of upcoming steps. In this way, the operation can be considered as correct (automatically skipping steps that are no longer needed) if the user has already reached the expected future state. Given that tutorials can have a large number of steps, a maximum look-ahead window of 10 future steps was adopted as a practical trade-off between sufficient context for accurate step recognition and feedback generation, and the computational cost required to maintain system responsiveness.

If the operation also requires validating the mesh topology, the comparison proceeds through the Mesh Processor component; otherwise, the flow continues through the Recommender component.

D. Mesh Processor

Comparing two meshes effectively is a complex task since users might perform operations different from those described in the tutorial steps, but still achieve the expected mesh. For instance, a tall pillar can be created from a cube either by extruding its top face or by manually adding new vertices above the existing ones and then connecting them to the base. The Mesh Processor component allows two meshes to be compared and, given a predefined margin of error set during the tutorial creation, to determine if they are equivalent. The component was designed to be independent, and its logic was not integrated into the Tutorial component, in order to make it possible to replace it with more sophisticated algorithms in the future.

Upon receiving the two meshes, the component first compares the number of vertices and faces of the mesh currently manipulated by the user with those of the expected one. If differences are detected, the process terminates and an error is reported to the user. Otherwise, the component deepens the comparison by adopting a greedy nearest-neighbor vertex-matching approach to establish one-to-one correspondences between the two meshes based on minimum Euclidean distance. If, at any point, a match cannot be found for a vertex in the user's mesh, the component immediately assumes the meshes are different. Following the vertex verification, the faces are checked using a similar logic to avoid scenarios in which meshes have the same number and arrangement of vertices but a different arrangement of faces, even with the same number of faces (as shown in Fig. 3). Edges are not explicitly compared, since in polygonal meshes they are implicitly defined by the face connectivity: therefore, verifying vertex correspondences together with face adjacency is sufficient to detect any topological differences, while avoiding unnecessary computational overhead.

E. Recommender

This component continuously monitors the learner's progress during Blender usage to keep the learner profile

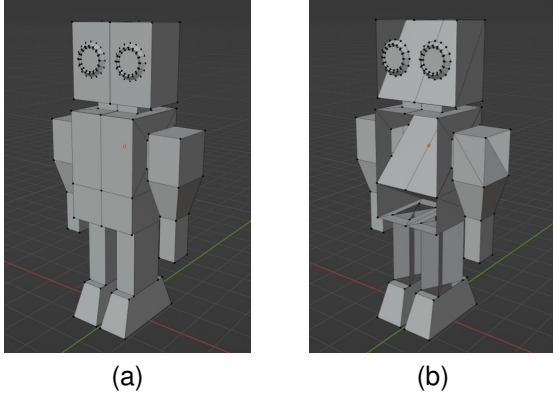


Fig. 3. Comparison of two meshes, (a) and (b), that have the same number of vertices, faces, and vertex arrangement but different arrangement of faces.

updated. Based on this profile, it detects observed difficulties to deliver immediate and continuous feedback as well as personalized tutorial recommendations using the CBF approach.

As shown in Fig. 1, the feedback is referred to as immediate and continuous because it updates promptly after each learner action, providing support throughout the entire activity rather than upon completion.

Once the tutorial has completed, the component produces a ranking of the most useful tutorials to recommend, based on the current learner profile. It is worth noticing that this approach requires a minimum amount of interaction to identify difficulties and update the learner profile. Indeed, when a learner performs a tutorial correctly without encountering difficulties, the system has insufficient evidence to characterize the learner's needs and, consequently, is not able to generate recommendations. In the current configuration, the two most relevant tutorials are passed to the UI component to be displayed on a dedicated panel in the Blender UI.

In addition to the recommended tutorials, this component also provides the UI component with the learner's two most significant difficulties, identified as the operations that were performed incorrectly most frequently during the execution of the tutorial. This information is easily retrieved by analyzing the learner profile and is displayed in the Blender UI (via the UI component) to explain the rationale behind the two recommended tutorials.

More details on learner profiling and the method used for generating recommendations are provided in Section V.

F. Cache

This component is responsible for storing ready-to-use information about the 3D objects and Blender's modifiers³ active in the scene. In this way, the data needed to validate the tutorial steps are directly available and ready to use without requiring further calculations or additional overhead due to the heterogeneity of data retrieval and data structures used internally by Blender. This is particularly important for Blender's modifiers, as they are crucial modeling operations that procedurally change an object's geometry or properties

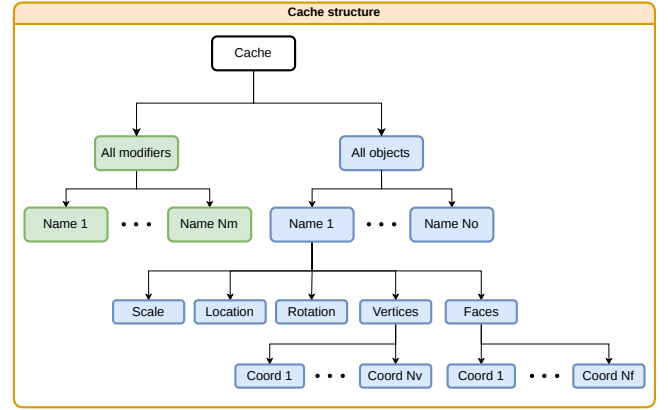


Fig. 4. Hierarchical structure used by the Cache component.

without permanently altering the original mesh. Fig. 4 shows the hierarchical organization of stored information.

Given that the primary focus of this work is on 3D modeling, this component stores, at present, only a sufficient amount of modifier-related information to support general modeling exercises, not fully covering workflows that rely exclusively on modifiers. In contrast, the information stored for objects is considerably more detailed. Each object is uniquely identified by its name (an approach that is robust since Blender does not allow objects with duplicate names) and stores metadata such as Scale, Location, and Rotation, as well as geometric information regarding its Vertices and Faces. By storing the state of objects, the Cache component guarantees that, whenever a new operation is detected, the system can use the cached information together with the current scene data to perform a reliable state comparison. This property is particularly crucial in operations that involve the addition or deletion of objects, since it is not possible to directly query Blender on objects that are recently added/removed due to known limitations of Blender's API. The Cache component also exposes several functionalities that allow manipulation of the stored data: these include the ability to save or retrieve information about a specific object, modify attributes of objects or modifiers, and perform state updates

G. Utils

This component provides auxiliary functionalities that support the functioning of other components. It offers general-purpose functions for retrieving information about 3D objects in the Blender scene (e.g., current number of objects, their names, the number of vertices and faces), handling data preparation for input/output operations (e.g., communicating the user which is the last correct operation recognized by the system and providing the immediate and continuous feedback), and managing the interaction with the user interface (e.g., updating the textual feedback displayed in Blender's UI). The importance of this component lies in the fact that it consistently retrieves the most up-to-date state of the scene. Therefore, all components invoking it can rely on fresh and ac-

³Blender's modifiers: <https://tinyurl.com/mvuzc5cr>

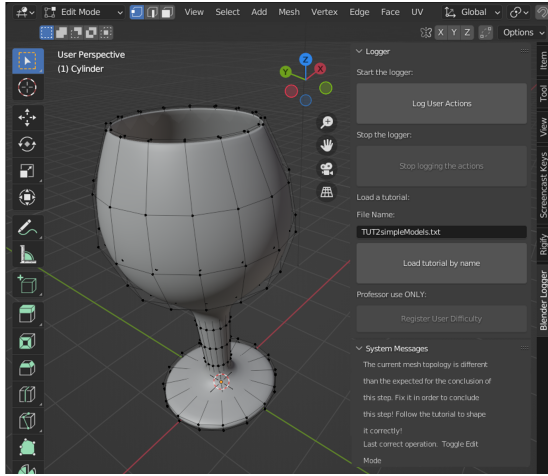


Fig. 5. Panel integrated in the Blender GUI to interact with the system. The Logger section shows the name of the tutorial being followed by the learner. The System Message section provides immediate feedback on performed operations (in this case, informs about a discrepancy in the mesh topology).

curate information, thus avoiding the propagation of outdated data across the system.

H. UI

This is the component responsible for rendering all graphical elements to be presented to the user when operating with the system. As illustrated in Fig. 5, it is implemented as a panel integrated into Blender’s interface. The panel is divided into two main sections.

The *Logger* section hosts the functionalities related to recording and loading tutorials, as well as manually flagging learner difficulties. Two buttons allow the user to activate/terminate the Log mode. When activated by pressing the *Log user actions* button, the system continuously records all operations performed by the user until the *Stop logging actions* button is pressed. The same section also allows the user to load a tutorial by entering its name. Once a valid tutorial name is provided, the system triggers the internal routines that prepare and initialize it to operate in Tutorial mode. Finally, a *Register user difficulty* button enables instructors to manually flag when a learner is unable to complete a tutorial step independently and requires direct assistance. More specifically, when a learner gets completely stuck, the instructor typically steps in to demonstrate the correct action. However, without manual logging, the system would falsely record the demonstration as a successful, error-free attempt by the learner. Using this button before assisting the learner ensures the failure is accurately logged, i.e., an error is reported even if the step’s actions are executed perfectly, thus maintaining the integrity of the learner profile.

The *System Messages* section is dedicated to providing immediate and continuous feedback to the learner on performed operations as well as reporting the final recommendations. Feedback messages vary in both type and format depending on the learner’s actions and tutorial progress. As an example, the message shown in Fig. 5 invites the learner to correct a discrepancy in the mesh topology by checking the expected

System Messages	Descriptions
<i>There are 4 faces missing in this object in order to conclude this step. Follow the tutorial to add them at the correct location!</i>	Message for a count discrepancy detected in Edit Mode (missing faces).
<i>Wrong operation! Expected operation: ‘Rename’ but got: ‘Toggle Edit Mode’</i>	Message reporting a discrepancy detected in the last performed operation.
<i>Correct operation. Your current progress: 2.6%. Next step: Perform the following operation: Select Face Mode on object ‘Cube’</i>	Message confirming a correct operation, showing progression and preview of the next step.
<i>Tutorial stopped early. Warning: the system could not gather enough information to make a recommendation nor identify your difficulties. Please, complete more tutorials!</i>	Message indicating that the current tutorial has been prematurely interrupted and that insufficient evidence is available for generating recommendations; the user is encouraged to complete more tutorials.
<i>Tutorial stopped early. Difficulties identified in the following operations: [‘(De)select All’, ‘Extrude Region and Move’] Tutorial recommended [‘TUTSimpleStairs.txt’, ‘TUTRing.txt’]</i>	Message indicating that the current tutorial has been prematurely interrupted; the system stops all ongoing profile updates and difficulty calculations, and generates the final recommendations, including the learner’s most problematic operations and the corresponding recommended tutorials.

TABLE I
EXAMPLES OF SYSTEM MESSAGES

mesh shape shown in the tutorial. Upon tutorial termination, this section outlines the learner’s most critical difficulties alongside recommended tutorials. This visualization allows learners to: i) understand the rationale underlying tutorial recommendations, and ii) become aware of their most frequent errors. Examples of system messages are reported in Table I.

V. RECOMMENDER ALGORITHM

Unlike traditional recommender system domains, such as movies, music, or e-commerce, where user preferences are often explicit and the items to be recommended are relatively homogeneous and predefined, in this work, the focus is on tutorials composed of sequences of heterogeneous 3D operations. Each tutorial, although referring to a particular topic, contains an implicit pathway of operations that learners must execute correctly to achieve a given result. Moreover, as said in the context of 3D modeling, there could be many different combinations of operations that yield the same final result. For this reason, the algorithm was designed to be diagnostic- rather than preference-driven. Thus, it continuously logs actions performed by the learner and constructs a dynamic profile, which is then used to recommend tutorials that address the learner’s specific weaknesses. In this scenario, content-based profiling in a vector space combined with cosine similarity and relevance feedback provide a way to estimate the proximity between learner’s needs and candidate tutorials, dynamically adapting the model to observed performance.

Below are provided the details of the methodologies devised to translate log-based tutorials in a vector space using the TF-IDF, update the learner profile in the same vector space by leveraging a revised version of the Rocchio [41] algorithm, and finally generate recommendations using a score calculated as the cosine similarity distance between vectors.

A. Tutorial Representation

A common approach in content-based recommender systems is to represent items as textual documents and map them into a vector space using TF-IDF, where each dimension corresponds to a term. In this work, since tutorials do not contain words but, rather, a sequence of Blender operations, they were modeled as “documents” whose terms are operation names instead of words.

Defining the vocabulary of all distinct operations observed in the tutorial library as $V = \{o_1, o_2, \dots, o_m\}$, a tutorial d can be represented as a vector $\mathbf{t}_d \in \mathbb{R}^m$ whose components are computed using TF-IDF as suggested in [36]. More specifically, let $f_{i,d}$ be the number of times operation o_i appears in tutorial d . The normalized term frequency $tf_{i,d}$ can be computed as follows:

$$tf_{i,d} = \frac{f_{i,d}}{\max_j f_{j,d}} \quad (1)$$

where $\max_j f_{j,d}$ is the maximum number of times any j -th operation appears in d .

To mitigate the operations that appear in many tutorials (thus reducing their relevance), the $tf_{i,d}$ is combined in the computation of the weights with the measure of inverse document frequency idf_i , computed as:

$$idf_i = \log \frac{N}{n_i} \quad (2)$$

where N is the total number of tutorials and n_i is the number of tutorials containing operation o_i .

The final weight $w_{i,d}$ is computed as $w_{i,d} = tf_{i,d} \cdot idf_i$ and the tutorial vector is expressed as $\mathbf{t}_d = (w_{1,d}, w_{2,d}, \dots, w_{m,d})$

The proposed weighting scheme emphasizes operations that are frequent within a tutorial but rare across the corpus, making them discriminative for matching. All tutorial vectors are precomputed and stored in a file that is loaded when the system is started.

B. Learner Profile Representation

The learner profile P is maintained in the same m -dimensional space ($\mathbf{P} \in \mathbb{R}^m$) used for representing tutorials, in which each dimension corresponds to a Blender operation. The profile is initialized as $\vec{P} = \vec{0}$. A positive value p_i encodes evidence of difficulty for operation o_i , while zero indicates no evidence yet. The profile is updated incrementally during interaction and normalized only at recommendation time. The Rocchio algorithm was selected to continuously update the learner profile in the vector space, moving it toward operations the learner finds difficult and away from those already mastered. The ability of this algorithm to update weights incrementally and rapidly makes it particularly suitable in this scenario, since it converges toward a meaningful profile within a single interaction, thus producing stable recommendations [28].

The vector of the learner profile is updated using the following revised version of the Rocchio formulation (proposed in [29]):

$$\vec{P}_{i+1} = \alpha \vec{P}_i + \beta \sum_{\vec{I}_j \in I^+} \vec{I}_j - \gamma \sum_{\vec{I}_k \in I^-} \vec{I}_k \quad (3)$$

where I^+ and I^- are the sets of items, i.e., Blender operations, considered relevant and non-relevant, respectively (in this case, a relevant operation refers to an operation the learner struggled with, whereas a non-relevant one is an operation correctly executed during the tutorials), $\vec{P}_{i+1}$ is the updated profile vector, whereas $\vec{P}_i$ is the current profile vector; α , β and γ are parameters that control the influence of the current profile, the relevant documents, and the non-relevant documents in the computation of the updated profile. In this work, the following values were used: $\alpha = 1.0$, $\beta = 0.5$, and $\gamma = 0.5$ to keep equal strength for positive (errors) and negative (successes) evidence (β and γ) while fully retaining history (α).

C. Continuous Relevance Feedback

In order to compute the complete item vectors I_j and I_k required to update the profile vector according to Eq. 3, a methodology was introduced that dynamically builds the profile as the learner progresses through the tutorial. More specifically, instead of computing Eq. 3 in batch form, incremental updates on the profile are considered. Each time the system classifies an attempt on operation o_i as wrong or correct, a one-hot vector $\mathbf{e}_i \in \mathbb{R}^m$ (i.e., 1 at index i , 0 elsewhere) is constructed and the incremental update is computed as:

- Wrong attempt on o_i : $\mathbf{P} \leftarrow \alpha \mathbf{P} + \beta \mathbf{e}_i$
- Correct attempt on o_i : $\mathbf{P} \leftarrow \alpha \mathbf{P} - \gamma \mathbf{e}_i$

This approach is algebraically equivalent to Rocchio with singleton sets $I^+ = \{\mathbf{e}_i\}$ or $I^- = \{\mathbf{e}_i\}$ accumulated over the session.

The batch formulation in Eq. 3 is therefore recovered at the end of the tutorial (or when the system is manually stopped) by summing all the incremental contributions. Finally, the following operations are performed before computing recommendations:

- Clipping at zero values: $\mathbf{P}_i \leftarrow \max(P_i, 0)$
- Normalization: $\hat{P} = \frac{\mathbf{P}}{\|\mathbf{P}\|}$ if $P \neq 0$

D. Recommendations Generation

To generate recommendations, the system calculates the cosine similarity between the normalized learner profile $\hat{P}$ and all available tutorial vectors $\vec{T}_d$:

$$\text{score}(d) = \frac{\hat{P} \cdot \vec{T}_d}{\|\hat{P}\| \|\vec{T}_d\|}$$

Tutorials are then ranked by their similarity scores, and the top-2 tutorials are recommended, explicitly excluding the current tutorial (to avoid recommending the same tutorial immediately again).

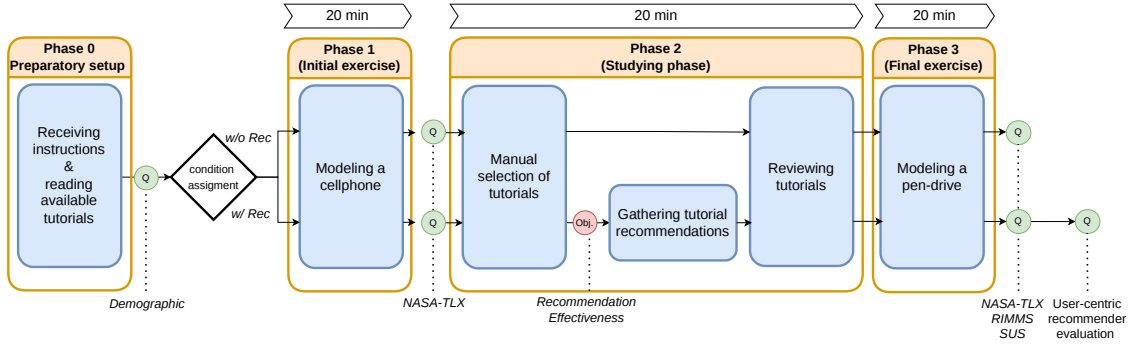


Fig. 6. Flow diagram of the experimental phases and evaluation protocol.

VI. EXPERIMENTAL EVALUATION

As stated before, the proposed ERS was designed to assist instructors during the laboratory sessions of a 3D modeling course like those taught at the authors' university. Building on this objective, an experimental evaluation was conducted to investigate the effects of using the proposed ERS to support hands-on activities comparable to those involved in such courses. In particular, the study aims to address the following research questions:

- RQ1: How *effective* is the ERS in the considered scenario?
- RQ2: What impact does a studying phase supported by the proposed ERS have on *learners' motivation* toward learning materials and the *perceived workload* when performing subsequent 3D modeling exercises?
- RQ3: How *usable* and *satisfactory* is the ERS from the learners' perspective?

A. Participants

The user study involved 26 participants (15 males and 11 females) aged between 22 and 29 ($M = 25.46$, $SD = 2.44$) and randomly recruited among students at the authors' university who have attended at least a CG course that involves the use of Blender for 3D modeling. According to the results of a demographic questionnaire collected at the beginning of the experiment, participants reported that they use Blender every day or at least once a week (69.24%). No participants identified as complete novices, i.e., "Never" used Blender. Moreover, 76.92% of the participants rated their familiarity with Blender tutorials as a 4 or 5 on a 5-point Likert scale from 1 "Not at all" to 5 "Very familiar". No participants identified themselves as "Not familiar at all" with Blender tutorials.

B. Study design

The study design (depicted in Fig. 6) follows a between-subjects approach to avoid carryover effects. More specifically, each participant was assigned to one of the following conditions: operating Blender i) with the recommender system running in the background to actively monitor activities and provide final recommendations (in the following referred to as the *w/ Rec* condition), or ii) without the recommender system (*w/o Rec* condition), thus resembling the hands-on activities

of the actual introductory course on 3D modeling considered as case study.

To replicate the workflow occurring in the laboratory sessions of that course, the study is organized into the following three phases (i.e., initial exercise, study phase, final exercise), preceded by a preparatory setup. Each phase has a maximum duration of 20 minutes, so that the entire experimental session, including questionnaires, fits into a time slot of roughly one and a half hours.

All the experiments were conducted with the same version of Blender adopted in the course in order to avoid potential bias related to the Blender GUI used.

1) *Preparatory setup phase*: During this phase, the participants were briefly briefed the study's structure, the basic instructions on how to operate the system and the conditions under which an operator (acting as a course instructor) could intervene during the experiment. More specifically, participants in both groups were expected to proceed autonomously, following the provided instructions. However, when experiencing critical bottlenecks, i.e., impossibility to proceed (e.g., they were not able to locate a required modeling tool or reproduce the intended geometry) during any phase of the study, they may ask the operator for help. In this case, the operator briefly diagnoses the situation, logs the request for support, and provides instructions to allow the participant to continue the experiment. The operator limited the intervention to short clarifications aimed solely at unblocking the participant, avoiding to provide direct solutions to the modeling task in general. This intervention mirrors the type of assistance provided, in actual laboratory settings, by the instructors. Typically, they observe the current state of a learner's work and attempt to correct errors by inferring the process that led to them, without knowing the exact steps performed by the learner, as they are simultaneously monitoring several learners. The possibility to request support was also introduced to prevent participants from remaining blocked for a prolonged period of time, and is essential, in this experimental setting, since it allows the participants to be exposed to a broader range of modeling operations and techniques, thereby increasing the likelihood of encountering different types of difficulties that the study aims to observe. The operator's oversight guaranteed that guidance remained consistent throughout the study, thereby ensuring the comparability of the measurements collected from different

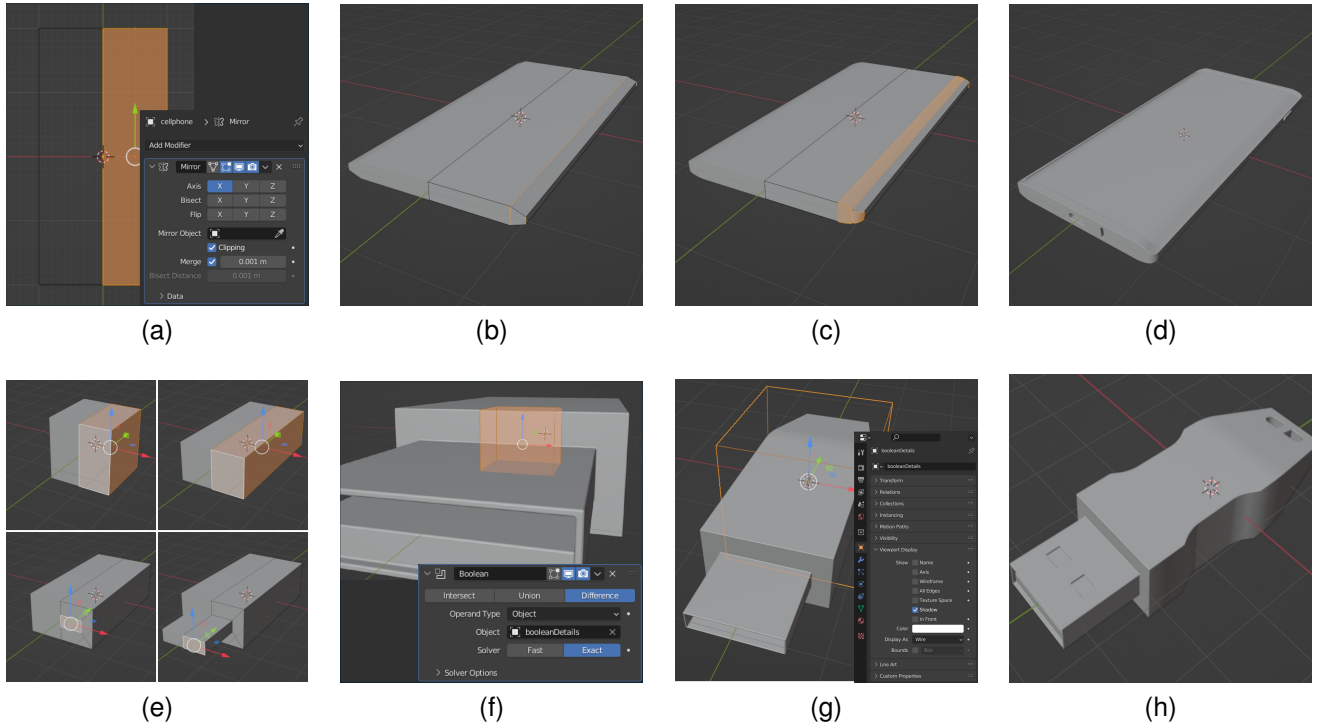


Fig. 7. Overview of the 3D modeling tasks involved in the experimental sessions. The first row (a–d) illustrates the modeling steps for the object assigned in Phase 1, i.e., a smartphone, whereas the second row (e–h) refers to the object modeled in Phase 3, i.e., a pen-drive. The screenshots represent a selection of the operations performed in Blender by the participants: (a) application of the Mirror modifier; (b) Loop cut creation and Edge Loop selection; (c) Bevel operation; (d) final model for Phase 1; (e) geometric transformations on mesh topologies and Extrude operation; (f) application of the Boolean modifier; (g) adjustment of rendering parameters via viewport display settings; (h) final model for Phase 3.

participants. Assistance was available to participants in both experimental conditions to ensure a balanced comparison and to maintain procedural consistency across the two groups.

During the preparatory phase, the operator also illustrated the tutorials that learners could use as study materials during the experiment, thus imitating the knowledge base traditionally available to learners of a course like the one considered as case study. The library included 19 tutorials spanning multiple modeling techniques covered in the syllabus of the course. **The material and texts of the tutorials used during the experiment are available for download at <https://tinyurl.com/2zzzf2ns>.**

Finally, participants were asked to complete the short questionnaire to collect demographic data, their familiarity with Blender, and information about their last participation in a CG course based on 3D modeling with Blender. Depending on their reported familiarity with Blender and the use of tutorials for 3D modeling, participants were assigned to one of the groups, aiming to balance the distribution of participants' skills between the two conditions.

2) Phase 1: The main goal of Phase 1 (initial exercise) is to establish a first approximation of each participant's modeling skills and difficulties. Moreover, this phase also provides a baseline measure of the workload experienced by the participant while performing a 3D modeling task. To this aim, the participants were asked to complete a modeling exercise following textual instructions provided on a separate sheet. The exercise required participants to model a simple smartphone (as shown in Fig. 7a) and is representative of the tasks

typically assigned in the considered introductory course. More specifically, the exercise spans a broad range of operations, including: basic transformations (move, rotate, scale), object manipulation (duplicate, delete), mesh editing (extrude, bevel, add edge loops), use of modifiers (e.g., boolean, solidify), and simple topology editing (e.g., loop selection).

3) Phase 2: During Phase 2 (study phase), participants consulted learning materials by working on a set of provided tutorials to address the difficulties encountered in the previous phase. This phase reproduces a common situation in laboratory-based learning, where learners consult tutorials or other documentation after having encountered difficulties in an exercise. In the w/o Rec condition, participants selected two tutorials they considered most useful from a list showing the tutorial names and covered operations. In the w/ Rec condition, the recommender system ranked tutorials available in the library based on the learner profile constructed after Phase 1 and suggested the two most relevant. Before viewing the system's suggestions, participants in this condition were also asked to choose their top-2 tutorials. In this way, it was possible to assess the effectiveness of the recommendations by directly comparing the recommended items with those selected independently by the users. The phase was limited to 20 minutes, thus participants who finished early could proceed immediately, while those still working at the time limit were instructed to stop and move to Phase 3.

4) Phase 3: This phase (final exercise) is designed to evaluate the effectiveness of the study phase. In an actual learning

scenario, learners typically apply newly acquired knowledge in a new exercise requiring similar modeling operations. To this end, participants are asked to complete an exercise that was structurally similar to the initial smartphone model and covers the same set of operations and concepts of Phase 1. In this case, the object to be modeled is the pen drive shown in Fig. 7b. As before, this phase was limited to 20 minutes.

C. Measures

During the experiments, both objective and subjective measurements were collected. Regarding the objective measurements, the following metrics were adopted:

- *Completion time* for the initial and final exercises (T_1 and T_2).
- *Completion percentage* of the two exercises at the end of Phase 1 ($\%Comp_1$) and Phase 3 ($\%Comp_2$).
- *Number of hints* requested by participants during the two exercises ($Hints_1$ and $Hints_2$).
- *Recommendation effectiveness* of the ERS, evaluated only for the participants in the w/ Rec group using the metrics reported below:
 - *Precision@2* measures the proportion of the top-2 recommended tutorials that are relevant to the participants (i.e., included in the participants' top-2 preferences).
 - *Recall@2* measures the proportion of relevant tutorials selected by the participants that are successfully retrieved within the top-2 recommendations.
 - *F1-score* calculated on Precision@2 and Recall@2.
 - *HitRate@2* indicates whether at least one relevant tutorial appears among the top-2 recommendations.
 - *Normalized Discounted Cumulative Gain (NDCG@2)* evaluates the quality of the ranking by assigning higher importance to relevant tutorials appearing in higher positions of the recommendation list; in the current evaluation, graded relevance is derived from the rankings provided by the participants.
 - *Mean Reciprocal Rank (MRR)* measures the average reciprocal position of the first relevant tutorial across the recommendation list.

For what it concerns the subjective measurements, participants completed the NASA-TLX [42] to assess the subjective workload associated with completing the exercises in Phase 1 and Phase 3, while the Reduced Instructional Materials Motivation Survey (RIMMS) [43] captured participants' motivation toward the learning materials. The System Usability Scale (SUS) [44] measured the usability of the tutorial selection procedure in Phase 2, whether automatic (w/ Rec condition, based on the profile built in Phase 1) or manual (w/o Rec condition, based on participants' perceived difficulties in Phase 1). Finally, the User-centric Recommender Evaluation Questionnaire [45] was administered only to participants in the w/ Rec condition after Phase 3, to evaluate the perceived effectiveness of the ERS.

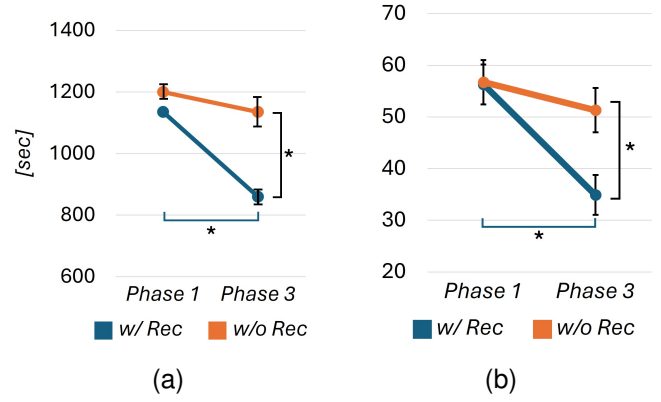


Fig. 8. Analysis of a) completion time and b) subjective workload across experimental phases. Error bars represent the standard deviation. Asterisks indicate significant differences ($p < .05$): vertical brackets denote simple condition effects (between-groups), whereas horizontal bracket denotes simple time effects (within-group).

VII. RESULTS

This section presents the results obtained by participants in the two conditions (w/ Rec vs. w/o Rec) and compares metrics after Phase 1 and Phase 3 (when appropriate).

To statistically analyze the impact of the ERS on completion time, completion percentage, number of hints requested, and subjective workload across experimental phases, a 2×2 mixed-design ANOVA was conducted with condition (w/ Rec vs. w/o Rec) as the between-subjects factor and phase (Phase 1 vs. Phase 3) as the within-subject factor.

Following the primary analysis, simple main effects and differences between conditions for the remaining metrics were analyzed. More specifically, normality was first assessed using the Shapiro–Wilk test, whereas homogeneity of variances was evaluated using Levene's test. When both assumptions were met, Student's t-test was applied; otherwise, Welch's t-test or the Mann–Whitney U test was used, as appropriate. To account for the multiple comparisons in post-hoc analyses (i.e., simple main effects for completion time, completion percentage, hint requests, and subjective workload), the Bonferroni correction was applied to all reported p -values: accordingly, the significance level was adjusted to $\alpha' = \alpha/4$, setting the threshold for statistical significance at $p < .0125$. Effect sizes were reported as Cohen's d for parametric tests and as r for non-parametric tests. All tests were two-tailed, with a significance level set at $\alpha = .05$.

A. Completion Time

The analysis on completion time (summarized in Table II) revealed a significant main effect of condition, $F(1, 24) = 17.54, p < .001, \eta_p^2 = .42$, with participants in the w/ Rec group being generally faster than those in the w/o Rec group. A significant main effect of phase was also observed, $F(1, 24) = 20.84, p < .001, \eta_p^2 = .46$, indicating a generic improvement in speed as participants gained experience. The analysis also yielded a significant phase $\times$ condition interaction, $F(1, 24) = 8.03, p = .009, \eta_p^2 = .25$, suggesting that

TABLE II
SUMMARY OF TWO-WAY MIXED ANOVA STATISTICAL RESULTS FOR OBJECTIVE METRICS AND SUBJECTIVE WORKLOAD

Outcome Variable	Phase	Condition	M (SD)	Significant Results
Completion time	1	w/ Rec	1135.38 (29.19)	$MC : F(1, 24) = 17.54, p < .001, \eta_p^2 = .42$ $MP : F(1, 24) = 20.84, p < .001, \eta_p^2 = .46$ $P \times C : F(1, 24) = 8.03, p = .009, \eta_p^2 = .25$ $P3 : w/ \text{rec} < w/o \text{rec}, p = .001, d = .63$ $w/o \text{rec} : P1 > P3, p = .001, d = .62$
		w/o Rec	1200.00 (0.00)	
	3	w/ Rec	859.62 (65.54)	
		w/o Rec	1135.46 (30.59)	
Completion percentage	1	w/ Rec	86.97 (4.38)	$MC : F(1, 24) = 13.57, p = .001, \eta_p^2 = .36$ $MP : F(1, 24) = 25.02, p < .001, \eta_p^2 = .51$
		w/o Rec	67.06 (3.87)	
	3	w/ Rec	98.82 (1.18)	
		w/o Rec	86.86 (4.67)	
Number of hints	1	w/ Rec	0.08 (0.08)	$MC : F(1, 24) = 24.34, p < .001, \eta_p^2 = .42$
		w/o Rec	1.23 (0.30)	
	3	w/ Rec	0.00 (0.00)	
		w/o Rec	0.69 (0.17)	
Subjective workload	1	w/ rec	56.31 (3.40)	$MC : F(1, 24) = 3.56, p = .072, \eta_p^2 = .13$ $MP : F(1, 24) = 18.30, p < .001, \eta_p^2 = .42$ $P \times C : F(1, 24) = 8.58, p = .007, \eta_p^2 = .25$ $P3 : w/ \text{rec} < w/o \text{rec}, p = .009, d = 1.11$ $w/ \text{rec} : P1 > P3, p < .001, d = 1.61$
		w/o Rec	56.72 (3.86)	
	3	w/ Rec	34.90 (3.93)	
		w/o Rec	51.31 (4.28)	

Note: MC = main effect of conditions; MP = main effect of phase; $P \times C$ = phase $\times$ condition interaction

the rate of improvement differed significantly between the two groups.

To further investigate this interaction, simple main effects were analyzed. After Phase 3, the gap between groups significantly increased. The w/ Rec group reached a mean completion time of 859.62s, whereas the w/o Rec remained significantly slower at 1135.46s ($p = .001, d = 0.63$). Moreover, a significant simple time effect was also observed for the w/o Rec group, which showed a reduction in completion time between Phase 1 and Phase 3 ($p = .001, d = 0.62$).

The progression of this metric across experimental phases and the corresponding statistically significant differences between and within groups are illustrated in Fig. 8a; more specifically, the figure highlights the significant reduction in completion time for the w/ Rec group, and reports the superior performance in terms of completion time achieved by the w/ Rec group with respect to the w/o Rec group in Phase 3.

B. Completion Percentage

For what it concerns completion percentage, the analysis revealed no significant phase $\times$ condition interaction, $F(1, 24) = 1.58, p = .221, \eta_p^2 = .06$. However, significant main effects were observed for both factors. A significant main effect of phase, $F(1, 24) = 25.02, p < .001, \eta_p^2 = .51$, confirmed that participants significantly increased their completion percentage over time, regardless of the condition. Moreover, a significant main effect of condition was found, $F(1, 24) = 13.57, p = .001, \eta_p^2 = .36$. As detailed in Table II, the w/ Rec group maintained a consistently higher completion rate regardless of the phase compared to the w/o Rec group.

C. Number of Hints

Regarding the number of hints requested by the participants, the analysis revealed a non-significant phase $\times$ condition

interaction, $F(1, 24) = 1.83, p = .189, \eta_p^2 = .07$, and a main effect of phase that approached but did not reach significance, $F(1, 24) = 3.25, p = .084, \eta_p^2 = .12$. However, a highly significant main effect of condition was observed, $F(1, 24) = 24.34, p < .001, \eta_p^2 = .50$. Participants in the w/ Rec group required no assistance during the whole experiment. In contrast, participants in the w/o Rec group relied significantly more on hints (as detailed in Table II).

D. Subjective Workload (NASA-TLX)

The analysis revealed a significant main effect of phase, $F(1, 25) = 18.30, p < .001, \eta_p^2 = .42$, indicating a general reduction in workload as participants progressed through the task regardless of the condition. The main effect of condition approached significance, $F(1, 24) = 3.56, p = .071, \eta_p^2 = .13$, suggesting an overall trend toward lower workload in the w/ Rec group. A significant phase $\times$ condition interaction was observed, $F(1, 25) = 8.58, p = .007, \eta_p^2 = .26$. This interaction indicates that the reduction in subjective workload over time followed significantly different trajectories depending on the experimental condition, as depicted in Fig. 8b.

To decompose this interaction, simple main effects were analyzed. After Phase 1 there was no significant difference in NASA-TLX scores between the two groups ($p = .937$), with both groups reporting high initial load ($M_{w/Rec} = 56.31$ vs. $M_{w/oRec} = 56.72$). After Phase 3, a significant difference emerged ($p = .009$) representing a large effect size ($d = 1.11$). The w/o Rec group experienced only a marginal decrease in load ($M = 51.31$), whereas the group supported by the ERS (w/ Rec) showed a substantial and statistically significant reduction between Phase 1 and Phase 3 ($p < .001, d = 1.61$).

E. Motivation Toward the Learning Materials (RIMMS)

Fig. 9 shows the overall scores computed for each dimension. In particular, overall scores were calculated by averaging

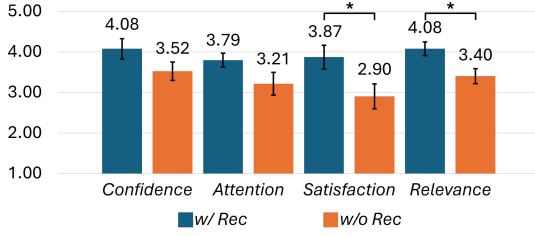


Fig. 9. Subjective results concerning motivation toward the learning materials based on RIMMS [43]. Error bars represent the standard deviation. Asterisks indicate significant differences ($p < .05$).

all statement values corresponding to each specific dimension. Significant differences were observed only for two dimensions, i.e., Satisfaction and Relevance. The w/ Rec group reported significantly higher overall Satisfaction ($M = 3.87$) compared to the w/o Rec group ($M = 2.90$, $p = .032$, $d = 0.87$). More specifically, analyzing the individual statements it was observed that participants in the w/ Rec group expressed a higher desire to learn more about the topic ($M = 3.85$ vs $M = 2.86$; $p = .036$, $d = 0.86$) and perceived the exercises as better designed ($M = 4.15$ vs $M = 3.14$; $p = .015$, $d = 1.00$). Concerning Relevance, a robust difference was found in the overall score ($p = .013$, $d = 1.03$), with the w/ Rec group ($M = 4.08$) perceiving the materials as more valuable than the w/o Rec group ($M = 3.40$). Significant advantages were noted regarding the worth of the content ($M = 4.15$ vs $M = 3.21$; $p = .016$, $d = 1.00$) and its perceived future utility ($M = 3.92$ vs $M = 2.93$; $p = .008$, $d = 0.53$).

Although the overall scores for Confidence and Attention dimensions did not reach the significance threshold ($p = .094$ and $p = .280$, respectively), the analysis of the individual statements revealed that the w/ Rec group felt significantly more confident due to the organization of the content ($M = 4.15$ vs $M = 3.14$; $p = .014$, $d = 0.48$). Additionally, the way information was arranged showed a marginal trend in favor of the w/ rec group for maintaining Attention ($M = 3.77$ vs $M = 2.93$; $p = .085$).

F. Overall Usability (SUS)

The analysis revealed a statistically significant difference in the overall SUS score between the two conditions ($M = 78.65$ vs $M = 67.17$; $p = .035$, $d = 0.85$), as shown in Table III. According to the categorization given by the authors of [46], the score obtained by the w/ Rec group corresponds to a C grade, associated to the “Good” class in the adjective rating scale, whereas the w/o Rec rated the condition with a D grade, corresponding to the “Ok” class.

Analyzing the individual statements, participants in the w/ Rec group were significantly more likely to indicate they would use the system frequently ($M = 4.00$ vs. 2.93 ; $p = .008$, $d = 0.52$). Moreover, the various functions within the recommendation system were perceived as significantly better integrated compared to the manual selection experience ($M = 4.23$ vs. 3.21 ; $p = .009$, $d = 0.52$). Finally, the w/ Rec group found the experience significantly less cumbersome ($M = 1.31$ vs. 2.71 ; $p = .002$, $d = 1.34$) and felt significantly

TABLE III
SUBJECTIVE RESULTS CONCERNING USABILITY BASED ON SUS [44].

Statement	w/ Rec M (SD)	w/o Rec M (SD)	p (d)
I think that I would like to use this system frequently	4.00 (0.20)	2.93 (0.27)	.008 ($d=.52$)
I found the system unnecessarily complex	1.62 (0.21)	2.14 (0.25)	.127
I thought the system was easy to use	4.00 (0.25)	3.86 (0.29)	.830
I think that I would need the support of a technical person to be able to use this system	2.38 (0.37)	1.71 (0.30)	.155
I found the various functions in this system were well integrated	4.23 (0.20)	3.21 (0.24)	.009 ($d=.52$)
I thought there was too much inconsistency in this system	1.92 (0.33)	2.00 (0.26)	.685
I would imagine that most people would learn to use this system very quickly	3.85 (0.30)	3.64 (0.32)	.793
I found the system very cumbersome to use	1.31 (0.13)	2.71 (0.37)	.002 ($d=.34$)
I felt very confident using the system	4.15 (0.15)	3.14 (0.29)	.014 ($d=.48$)
I needed to learn a lot of things before I could get going with this system	1.54 (0.22)	1.36 (0.17)	.616
SUS Score	78.65 (2.53)	67.14 (4.42)	.035 ($d=.85$)
Grade	C	D	
Adjective rating	Good	Ok	

TABLE IV
OBJECTIVE RESULTS CONCERNING RECOMMENDATION EFFECTIVENESS

Metric	M (SD)	Metric	M (SD)
Precision@2	0.58 (0.05)	HitRate@2	1.00 (0.00)
Recall@2	0.58 (0.05)	NDCG@2	0.64 (0.06)
F1-score	0.58 (0.05)	MRR	0.88 (0.06)

more confident using the system ($M = 4.23$ vs. 3.21 ; $p = .014$, $d = 0.48$).

G. W/ Rec Group-specific Metrics (Recommendation effectiveness metrics and user-centric recommender evaluation questionnaire)

As anticipated, to conclude the analysis, the effectiveness of the ERS was evaluated through a number of objective and subjective measurements collected only for participants in the w/ Rec group.

The aim of the objective metrics is to assess the accuracy, retrieval capability and ranking quality of the suggestions provided during the experimental tasks, focusing on a recommendation window of $k = 2$. More specifically, the evaluation involved comparing a ranked list of the two tutorials proposed by the system against a ranked list of two tutorials identified as relevant by the participant. This ranking-to-ranking comparison served as the basis for computing the accuracy and quality metrics detailed below.

As reported in Table IV, the proposed ERS demonstrated robust predictive capabilities and high alignment with learning needs. More specifically, the ERS achieved a Precision@2 of 0.58 (± 0.05) and a Recall@2 of 0.58 (± 0.05), resulting in a balanced F1-score of 0.58 (± 0.05). These values indicate that nearly half of the generated suggestions were directly relevant to the participants’ immediate needs, and the system successfully retrieved potentially relevant tutorials within the top-2 slots.

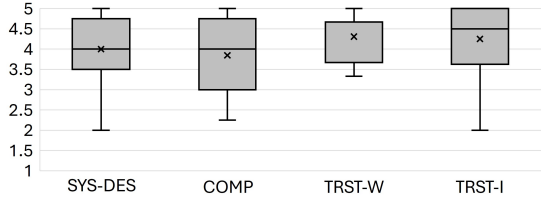


Fig. 10. Subjective results based on User-centric Recommender Evaluation Questionnaire [45]. Abbreviations: SYS-DES (Subjective perceptions of system-design features), COMP (Overall competence perceptions), TRST-W (Trustworthiness), and TRST-I (Trusting intentions). The “x” symbol represents the mean value, while the central horizontal line indicates the median. The whiskers extend to the minimum and maximum observed values.

In addition, the ERS achieved a $\text{HitRate}@2$ of $1.00 (\pm 0.00)$, meaning that in every instance, the system provided at least one relevant tutorial within the top-2 recommendations. The quality of the ranking was further validated by an $\text{NDCG}@2$ of $0.64 (\pm 0.06)$ and an MRR of $0.88 (\pm 0.09)$. The high MRR score suggests that relevant items were consistently placed in the first position of the list.

The findings related to the User-centric Recommender Evaluation Questionnaire are summarized in Fig. 10. Overall scores were calculated by first inverting the scores of individual statements in the case of reverse-scaled items, and subsequently averaging all statement values corresponding to each specific dimension. Starting from the highest scores assigned to the individual statements of the “Perceptions of system-design features” dimension, it is possible to notice that participants reported a clear understanding of the recommendation logic ($M = 3.92$) and highly valued the quality of the suggestions provided ($M = 4.08$). Moving to the “Overall competence perceptions” dimensions, the ERS was also perceived as highly effective in helping users find the tutorials they needed ($M = 4.46$). Participants felt very confident that the recommendations were the “best choices” for them ($M = 4.31$). The reverse-scaled item regarding effort ($M = 1.85$) confirms that searching for tutorials required very low cognitive strain. Trustworthiness received the highest ratings in the questionnaire. Participants felt the ERS was highly trustworthy ($M = 4.62$) and specifically trusted the recommendations because they were consistent with their individual difficulty levels ($M = 4.46$). Overall satisfaction remained above the neutral value ($M = 3.85$). The positive experience translated into strong future intentions evaluated in the trusting intentions section. Indeed, participants expressed a high likelihood of using a similar ERS for future searches ($M = 4.08$) and believed that using the ERS again would help them find suitable tutorials more quickly ($M = 4.15$).

VIII. DISCUSSION

This section analyzes the experimental results in relation to the established research questions, exploring the effectiveness (RQ1), impact of the ERS in the study phase (RQ2), as well as the perceived satisfaction and usability (RQ3).

A. Technical and Functional Effectiveness (RQ1)

The data confirms that the devised ERS is a highly effective tool for providing targeted educational support in the considered context. The high values of $\text{HitRate}@2$ indicate that the algorithm’s recommendations consistently aligned with learning needs, effectively proposing tutorials needed by the participants. The reliability of the ERS is also supported by values of $\text{Recall}@2$, MRR and $\text{NDCG}@2$, indicating that not only were relevant tutorials retrieved, but they were consistently placed at the top of the recommendation list. The technical strengths shown by the proposed ERS translated directly into objective performance improvements, as confirmed, e.g., by the near-zero requests for hints in the w/ Rec group during the experiments. Looking at the remaining objective metrics, it is also possible to observe that, not only the ERS was useful in Phase 2 (where participants were asked to search for tutorials), but it also influenced the next phase. In particular, during Phase 3, participants in the w/ Rec group were able to complete the task significantly faster than those in the w/o Rec group, confirming the usefulness of the tutorials recommended in the previous phase. The significant interaction effect on completion time further suggests that the ERS accelerates the learning process, particularly the acquisition and application of the skills required to complete the assigned tasks, compared to manual tutorial selection.

Results related to the completion percentage highlight that the ERS allowed participants to complete a higher percentage of the required operations regardless of the phase. This result can be attributed to the system’s immediate and continuous feedback, which helped participants in the w/ Rec group immediately recognize and correct errors as they occurred, preventing mistakes from accumulating over time. In contrast, participants in the w/o Rec group were often unaware of issues until they became difficult to manage, making recovery more challenging not only for them, who were required to ask for support, but also for the instructor, who had to resolve multiple interconnected problems before the participants could continue with the exercise (i.e., the same issue observed during actual laboratory sessions). Moreover, in Phase 3, almost all participants in the w/ Rec group were able to complete the exercise. Although the w/o Rec group also improved over time, its completion rate by the end of the study was only comparable to that achieved by the w/ Rec group already in Phase 1.

B. Workload and Motivational Value (RQ2)

The results based on the NASA-TLX [42] and RIMMS [43] questionnaires suggest that the ERS had a significant impact on both the workload and motivation toward the instructional material. More specifically, although both groups began with high mental strain associated with the execution of the first exercise, the w/ Rec group reached a significantly more manageable level of workload by the last phase. The large interaction effect size suggests that, by offloading the effort of searching for tutorials, the system reduced participants’ workload as they were able to practice more tutorials specifically aligned with their difficulties. As discussed previously, the recommended

tutorials proved to be effective and were also perceived as such by the participants, probably making them feel more prepared and confident when dealing with the second modeling exercise.

The benefits observed in workload are closely related to the gains in motivation. The significantly higher scores in relevance and satisfaction suggest that participants perceived the learning materials as more useful when they were recommended by the proposed ERS. This result is probably connected to the complexity of many 3D modeling techniques (included in the two exercises), which are often composed of multiple operations or can be executed through different procedural variations. As a consequence, when participants relied only on the name of a technique to search for tutorials (that is the condition in which the w/o Rec group operated), they could select materials that addressed the general technique but only partially covered the specific operations responsible for their difficulties. In contrast, the ERS was able to identify the specific challenges of the participants and recommend tutorials containing a larger number of the operations with which participants struggled. The higher alignment between the recommended content and the participants' actual needs increased both the perceived usefulness of the tutorials and the motivation to engage with them.

C. Usability and Satisfaction (RQ3)

From the learners' perspective, the ERS fundamentally improved the usability of the learning environment evaluated through the SUS questionnaire [44].

The gap between the SUS scores assigned by the w/ Rec and w/o Rec groups suggests that manually selecting tutorials was perceived as less usable than receiving recommendations directly from the proposed system. This finding is particularly noteworthy considering that manual tutorial selection was a familiar and commonly adopted process for the participants, whereas the recommender system was completely new to them. Moreover, participants performed the manual search within a relatively small tutorial repository, much smaller than those typically available in actual educational settings. More specifically, the analysis of the individual SUS statements revealed that participants found the experience significantly less cumbersome, felt more confident in their actions, and perceived the various functions involved in the tutorial selection process as better integrated.

The user-centric evaluation based on [45] reveals particularly high levels of perceived trustworthiness and recommendation effectiveness. Participants strongly trusted the ERS and considered the recommended tutorials highly consistent with their individual difficulties, suggesting that the system successfully provided personalized and reliable support. Moreover, users perceived the ERS as highly effective in helping them identify the tutorials they actually needed, while also reporting strong confidence that the recommended tutorials represented the best choices for their learning process. These findings are further reinforced by the very low effort score, indicating that the system significantly reduced the cognitive burden associated with manually searching for suitable tutorials.

IX. CONCLUSION AND FUTURE WORKS

This paper presented a novel, domain-specific ERS natively integrated into Blender to address the steep learning curve and complex workflows inherent in CG education. The technical core of the system leverages a content-based filtering strategy that utilizes a TF-IDF-based representation of tutorials and a revised Rocchio update mechanism for continuous learner profiling. This allows the system to identify specific operational difficulties and provide immediate and continuous feedback as well as context-aware tutorial recommendations.

The experimental evaluation involving 26 participants demonstrated that the ERS is highly effective, as confirmed by the positive results achieved across the objective metrics. In particular, participants supported by the proposed ERS completed assigned tasks significantly faster and with higher completion rates than the control group, effectively reducing the cumulative impact of modeling errors. In addition, the system significantly reduced the perceived workload while enhancing motivation and satisfaction, as the suggested materials were directly aligned with the immediate learning needs. The ERS was perceived as highly trustworthy and intuitive, achieving superior usability scores compared to manual tutorial selection, which represents the common practice in 3D modeling laboratory settings.

Despite these positive results, several directions remain open for future research. The present study focused on the design and evaluation of the system within a single introductory CG course; therefore, its applicability across different courses, instructors, learner populations, and educational contexts remains to be investigated [through additional experiments](#).

Moreover, although the underlying architecture can support different Blender-based exercises and learning resources, deploying it in new educational settings, including more advanced CG topics such as procedural geometry nodes, animation, and physics, would require the creation and configuration of a corresponding tutorial library, [as well as, where needed, domain-specific result-processing components to assess the correctness of outcomes that may be achieved through multiple valid workflows \(e.g., alternative animation keyframe configurations or geometry nodes architectures\)](#). Such components can be implemented as interchangeable modules, [similarly to the Mesh Processor used for 3D modeling, and therefore extend the existing ERS without changing its underlying architecture](#). Furthermore, extending the proposed approach to other software environments, such as Maya, would also require [adapting several components, e.g., to intercepting properly the user inputs and accessing the underlying data structures](#). Future studies could also explore the adoption of the proposed approach in educational domains beyond CG. Finally, long-term longitudinal studies could be conducted to assess the impact of integrated ERSs on knowledge retention.

REFERENCES

- [1] A. Hofstein and V. N. Lunetta, "The laboratory in science education: Foundations for the twenty-first century," *Science Education*, vol. 88, no. 1, pp. 28–54, 2004.
- [2] M. Abdulwahed and Z. K. Nagy, "Applying kolb's experiential learning cycle for laboratory education," *Journal of Engineering Education*, vol. 98, no. 3, pp. 283–294, 2009.

- [3] M. K. Seery, H. Y. Agustian, F. V. Christiansen, B. Gammelgaard, and R. H. Malm, "10 guiding principles for learning in the laboratory," *Chemistry Education Research and Practice*, vol. 25, no. 2, pp. 383–402, 2024.
- [4] A. van Leeuwen, J. Janssen, G. Erkens, and M. Brekelmans, "Teacher regulation of multiple computer-supported collaborating groups," *Computers in Human Behavior*, vol. 52, pp. 233–242, 2015.
- [5] R. Kasepalu, P. Chejara, L. P. Prieto, and T. Ley, "Studying teacher withness in the wild: Comparing a mirroring and an alerting & guiding dashboard for collaborative learning," *International Journal of Computer-Supported Collaborative Learning*, vol. 18, no. 4, pp. 575–606, 2023.
- [6] T. Hattori, R. Masuda, Y. Moritoh, Y. Imai, Y. Kawakami, and T. Tanaka, "Utilization of both free 3dcg software "blender" and 3d printing for early stem education," in *Proc. of IEEE International Conference on Teaching, Assessment, and Learning for Engineering*, 2020, pp. 879–882.
- [7] T. Suselo, B. C. Wünsche, and A. Luxton-Reilly, "Technologies and tools to support teaching and learning computer graphics: A literature review," in *Proc of Australasian Computing Education Conference*, 2019, p. 96–105.
- [8] F. Lamberti, G. Paravati, V. Gatteschi, A. Cannavo, and P. Montuschi, "Virtual character animation based on affordable motion capture and reconfigurable tangible interfaces," *IEEE Transactions on Visualization and Computer Graphics*, vol. 24, no. 5, pp. 1742–1755, 2017.
- [9] A. Sanna, F. Lamberti, G. Paravati, and C. Demartini, "Automatic assessment of 3d modeling exams," *IEEE Transactions on Learning Technologies*, vol. 5, no. 1, pp. 2–10, 2011.
- [10] R. Bodily and K. Verbert, "Review of research on student-facing learning analytics dashboards and educational recommender systems," *IEEE Transactions on Learning Technologies*, vol. 10, no. 4, pp. 405–418, 2017.
- [11] D. Herath, C. Dinuwan, C. Ihalagedara, and T. Ambegoda, "Enhancing educational outcomes through AI powered learning strategy recommendation system," *International Journal of Advanced Computer Science and Applications*, 2024.
- [12] E. al. Ipseeta Nanda, "ALBERT-based personalized educational recommender system: Enhancing students' learning outcomes in online learning," *International Journal on Recent and Innovation Trends in Computing and Communication*, 2023.
- [13] S. Souabi, A. Retbi, M. K. Idrissi, and S. Bennani, "Recommendation systems on e-learning and social learning: A systematic review," *Electronic Journal of e-Learning*, 2021.
- [14] G. Lampropoulos, "Recommender systems in education: A literature review and bibliometric analysis," *Advances in Mobile Learning Educational Research*, 2023.
- [15] A. Y. Q. Huang, O. H. T. Lu, and S. J. H. Yang, "Effects of artificial intelligence-enabled personalized recommendations on learners' learning engagement, motivation, and outcomes in a flipped classroom," *Computers & Education*, vol. 194, p. 104684, 2023.
- [16] N. Kamal, F. Sarker, A. Rahman, S. Hossain, and K. Mamun, "Recommender system in academic choices of higher education: A systematic review," *IEEE Access*, vol. 12, pp. 35 475–35 501, 2024.
- [17] M. Deschênes, "Recommender systems to support learners' agency in a learning context: A systematic review," *International Journal of Educational Technology in Higher Education*, vol. 17, 2020.
- [18] M. Dascalu, C. Bodea, M. N. Mihailescu, E.-A. Tanase, and P. O. Pablos, "Educational recommender systems and their application in lifelong learning," *Behaviour & Information Technology*, vol. 35, pp. 290 – 297, 2016.
- [19] K. Verbert, N. Manouselis, X. Ochoa, M. Wolpers, H. Drachsler, I. Bosnic, and E. Duval, "Context-aware recommender systems for learning: A survey and future challenges," *IEEE Transactions on Learning Technologies*, vol. 5, pp. 318–335, 2012.
- [20] S. Bhatt, K. Verbert, and W. V. den Noortgate, "Teacher-centric educational recommender systems in K12 practice: Usage and evaluation," *Heliyon*, vol. 11, 2025.
- [21] S. E. Ayman and A. A. E. Rejal, "Ontology and machine learning-based recommender system for teacher resource personalization," *Education Science and Management*, 2023.
- [22] H. Drachsler, K. Verbert, O. C. Santos, and N. Manouselis, "Panorama of recommender systems to support learning," in *Recommender systems handbook*. Springer, 2015, pp. 421–451.
- [23] N. Askarbekuly and I. Luković, "Learning outcomes, assessment, and evaluation in educational recommender systems: A systematic review," *arXiv preprint arXiv:2407.09500*, 2024.
- [24] A. C. Rivera, M. Tapia-Leon, and S. Lujan-Mora, "Recommendation systems in education: A systematic mapping study," in *Proc. of International conference on information technology & systems*. Springer, 2018, pp. 937–947.
- [25] I. Gula, O. Maznichenko, A. Kutsenko, A. Osadcha, and N. Kravchenko, "Methods of teaching graphic design in heis for art," *Journal of Curriculum and Teaching*, 2023.
- [26] F. Lamberti, A. Sanna, G. Paravati, and G. Carlevaris, "Automatic grading of 3d computer animation laboratory assignments," *IEEE Transactions on Learning Technologies*, vol. 7, no. 3, pp. 280–290, 2014.
- [27] T. Chlubna, M. Vlnas, T. Milet, and P. Zemčík, "Survey of foss 3d/2d graphics software Blender usage in science, academia, and industry," *The Visual Computer*, vol. 42, no. 1, p. 95, 2026.
- [28] G. Salton and C. Buckley, "Improving retrieval performance by relevance feedback," *Journal of the American Society for Information Science*, vol. 41, no. 4, pp. 288–297, 1990.
- [29] C. Wang, Y. Shen, H. Yang, and M. Guo, "Improving rocchio algorithm for updating user profile in recommender systems," in *Web Information Systems Engineering – WISE 2013*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 162–174.
- [30] M. C. Urdaneta-Ponte, A. Mendez-Zorrilla, and I. Oleagordia-Ruiz, "Recommendation systems for education: Systematic review," *Electronics*, vol. 10, no. 14, p. 1611, 2021.
- [31] L. Salau, M. Hamada, R. Prasad, M. Hassan, A. Mahendran, and Y. Watanobe, "State-of-the-art survey on deep learning-based recommender systems for e-learning," *Applied Sciences*, vol. 12, no. 23, p. 11996, 2022.
- [32] M. M. Afsar, T. Crump, and B. Far, "Reinforcement learning based recommender systems: A survey," *ACM Computing Surveys*, vol. 55, no. 7, pp. 1–38, 2022.
- [33] J. Jeevamol and V. Renumol, "An ontology-based hybrid e-learning content recommender system for alleviating the cold-start problem," *Education and Information Technologies*, vol. 26, no. 4, pp. 4993–5022, 2021.
- [34] J. Joy and R. V. G. Pillai, "Review and classification of content recommenders in e-learning environment," *Journal of King Saud University Computer and Information Sciences*, vol. 34, no. 9, pp. 7670–7685, 2022.
- [35] R. Burke, "Hybrid recommender systems: Survey and experiments," *User Modeling and User-adapted Interaction*, vol. 12, no. 4, pp. 331–370, 2002.
- [36] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions," *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 6, pp. 734–749, 2005.
- [37] L. Fernandez-Luque, R. Karlsen, and L. K. Vognild, "Challenges and opportunities of using recommender systems for personalized health education," *Medical Informatics in a United and Healthy Europe*, pp. 903–907, 2009.
- [38] F. L. Da Silva, B. K. Slodkowski, K. K. A. Da Silva, and S. C. Cazella, "A systematic literature review on educational recommender systems for teaching and learning: research trends, limitations and opportunities," *Education and Information Technologies*, vol. 28, no. 3, pp. 3289–3328, 2023.
- [39] C. De Medio, C. Limongelli, F. Sciarrone, and M. Temperini, "Moodlerec: A recommendation system for creating courses using the moodle e-learning platform," *Computers in Human Behavior*, vol. 104, p. 106168, 2020.
- [40] O. Mosendz, S. Borysova, K. Vitychynkina, A. Brovchenko, and O. Kovalchuk, "Experimental methods of teaching computer graphics and design: Impact on the creative potential of higher education students," *Salud, Ciencia y Tecnología-Serie de Conferencias*, vol. 4, p. 640, 2025.
- [41] J. J. Rocchio Jr, "Relevance feedback in information retrieval," *The SMART Retrieval System: Experiments in Automatic Document Processing*, 1971.
- [42] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (task load index): Results of empirical and theoretical research," in *Advances in psychology*. Elsevier, 1988, vol. 52, pp. 139–183.
- [43] N. Loorbach, O. Peters, J. Karreman, and M. Steehouder, "Validation of the instructional materials motivation survey (imms) in a self-directed instructional setting aimed at working with technology," *British Journal of Educational Technology*, vol. 46, no. 1, pp. 204–218, 2015.
- [44] J. Brooke et al., "Sus-a quick and dirty usability scale," *Usability Evaluation in Industry*, vol. 189, no. 194, pp. 4–7, 1996.
- [45] P. Pu, L. Chen, and R. Hu, "A user-centric evaluation framework for recommender systems," in *Proc. of the ACM conference on Recommender systems*, 2011, pp. 157–164.

- [46] B. Aaron, K. Philip, and M. James, “Determining what individual SUS scores mean: Adding an adjective rating scale,” *Journal of Usability Studies*, vol. 4, no. 3, pp. 114–123, 2009.



Alberto Cannavò is a fixed-term tenure-track assistant professor at the Department of Control and Computer Engineering, Politecnico di Torino, Turin, Italy. His current research interests include computer graphics, extended reality, and human-machine interaction. He received his Ph.D. degree in computer engineering from Politecnico di Torino. He is currently serving as Associate Editor of IEEE Transactions on Consumer Electronics. He is a Member of IEEE and the IEEE Consumer Technology Society. Contact him at alberto.cannavo@polito.it.



Roberta Macaluso is a Ph.D. student at Politecnico di Torino, Turin, Italy. Her current research interests include socially engaged virtual environments, eXtended Reality collaboration, and human-computer interaction. She received her M.Sc. degree in computer engineering from Politecnico di Torino. She is a Student Member of IEEE and the IEEE Consumer Technology Society. Contact her at roberta.macaluso@polito.it.



Pier Luigi Nakai Ricchetti is the Co-Founder and Chief Technology Officer of Navigare, São Paulo, Brazil, where he leads the development of artificial intelligence solutions for international trade and customs compliance. His current research interests include artificial intelligence, machine learning, large language models, knowledge-based reasoning, and intelligent decision-support systems. He received his M.Sc. degree in computer engineering from Politecnico di Torino, Turin, Italy. Contact him at ricchetti@nvgr.io.



Fabrizio Lamberti is a Full Professor at the Department of Control and Computer Engineering, Politecnico di Torino, Turin, Italy, where he leads the G^Raphics And INtelligent Systems group and the VR@POLITO lab. His current research interests include computer graphics, human-machine interaction, and intelligent systems. He received his Ph.D. degree in computer engineering from Politecnico di Torino. He is currently serving as Editor-in-Chief of IEEE Consumer Electronics Magazine and as Associate Editor of IEEE Transactions on Visu-

alization and Computer Graphics. He is the Vice President of Technical Activities of IEEE Consumer Technology Society, and Chair of IEEE Computer Society Italy Chapter. He is Senior Member of IEEE. Contact him at fabrizio.lamberti@polito.it.